VIKRAM KUMAR

Bokaro Steel City, Jharkhand - 829301 +91 9102430603
vikramkumarbcrec2027@gmail.com vikram-kumar-0b19a9248 prajapativikram

Professional Summary

Final-year B.Tech Computer Science and Engineering student with experience in Java, Spring Boot, REST API development, and full-stack applications. Built backend services and integrated frontend, backend, and PostgreSQL database components. Familiar with object-oriented programming, software development lifecycle practices, API testing, and collaborative application development.

Experience

CSIR-Central Mechanical Engineering Research Institute (CSIR-CMERI) **Sep 2026 – Sep 2026**
Research Intern *Durgapur, West Bengal*

- Analyzed trajectory data from exoskeleton-assisted path-following experiments for visually impaired individuals using Python and OpenCV.
- Implemented computer vision techniques for LED marker detection, coordinate extraction, and perspective transformation from experimental video frames.
- Visualized actual and reference trajectories using NumPy, Pandas, and Matplotlib to evaluate path-following accuracy.
- Performed trajectory error analysis to assess deviations between experimental and reference paths.

Infosys Springboard **Jan 2026 – Mar 2026**
Software Engineer Intern *Remote*

- Developed backend services using Java and Spring Boot for web-based business application workflows.
- Designed and implemented REST APIs to support application features and integration between services.
- Applied object-oriented principles to organize maintainable backend components.
- Collaborated on development activities using software development lifecycle practices.

Education

Dr. B. C. Roy Engineering College **Aug. 2023 – Jun. 2027**
Bachelor of Technology in Computer Science and Engineering
Current SGPA: 8.68/10 *Durgapur, WB*

S L Arya Inter College **May 2020 – May 2022**
Intermediate of Science – 4th District Merit
Grade: 93.6% *Bokaro, JH*

Saraswati Shishu Vidya Mandir **Apr. 2019 – Apr. 2020**
Secondary Education
Grade: 90.8% *Bokaro, JH*

Projects

GenAI Hub – Full-Stack AI SaaS Application **Nov 2025 – Dec 2025**
Tech Stack: React.js, Node.js, Express.js, PostgreSQL, Generative AI APIs

- Developed a full-stack web application for resume analysis, content generation, and PDF processing using React.js, Node.js, and Express.js.
- Integrated REST APIs to enable communication between frontend components, backend services, and Generative AI models.
- Utilized PostgreSQL (NeonDB) for database integration and supporting application workflows.

Intelligent Video-Based Accident Detection System **Jan 2026 – Feb 2026**
Tech Stack: Python, PyTorch, Swin Transformer, OpenCV, Streamlit

- Developed a video-based accident detection application using Python, PyTorch, and Swin Transformer for automated classification.
- Implemented video processing and frame extraction using OpenCV to identify potential accident events.
- Built an interactive Streamlit interface for video uploads, accident predictions, and visualization of detected frames.
- Achieved 94% classification accuracy with 94% weighted F1-score on the test dataset.

Technical Skills

Programming Languages: Java, Python, JavaScript, C, SQL
Backend & Frameworks: Spring Boot, Node.js, Express.js, React.js
Databases: MySQL, PostgreSQL (NeonDB), SQL
Web Technologies: HTML5, CSS3, REST APIs, API Integration
Developer Tools: Git, GitHub, VS Code, Postman, Jupyter Notebook
AI/ML Technologies: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Generative AI APIs
Core Concepts: Data Structures & Algorithms @VikramKr, Object-Oriented Programming (OOP) , SDLC, API Testing, Debugging

Achievements & Additional Information

- Languages: English (Fluent), Hindi (Native).
- Earned NPTEL Elite + Silver Certification (91%) in The Joy of Computing Using Python, IIT Madras (2025).
- Completed AI/ML training under the MSME Entrepreneurship-cum-Skill Development Programme at CTTC Bhubaneswar, Government of India (2026).
- Solved 300+ coding problems on LeetCode, strengthening problem-solving skills in Data Structures, Algorithms, and algorithmic optimization.